

The diagram illustrates a digital circuit for a 26-bit LPM counter. The main components and their connections are as follows:

- LPM_COUNTER**: A 26-bit counter with inputs `updown`, `clk_en`, and `cnt_en`. It is connected to `VCC` and `GND`. Its output `q[]` is connected to `CNT[25..0]`.
- 7474 D FLIP-FLOPS**: A D flip-flop with inputs `1PRN`, `1D`, `1CLRn`, `1CLK`, `2PRN`, `2D`, `2CLRn`, and `2CLK`. It has outputs `1Q`, `1Qn`, `2Q`, and `2Qn`. It is connected to `VCC` and `GND`.
- 74164 SHIFT REG.**: A shift register with inputs `A`, `B`, `CLRn`, and `CLK`. It has outputs `QA`, `QB`, `QC`, `QD`, `QE`, `QF`, `QG`, and `QH`. It is connected to `VCC` and `GND`.
- 74160 COUNTER**: A counter with inputs `LDN`, `A`, `B`, `C`, `D`, `ENT`, `ENP`, `CLRn`, and `CLK`. It has outputs `QA`, `QB`, `QC`, `QD`, and `RCO`. It is connected to `VCC` and `GND`.
- 74166 DECODER**: A decoder with inputs `A`, `B`, `C`, and `D`. It has outputs `Y0`, `Y1`, `Y2`, `Y3`, `Y4`, `Y5`, `Y6`, and `Y7`. It is connected to `VCC` and `GND`.

The circuit is connected to various pins and components:

- PIN_R8**: Connected to `CLOCK_50`.
- PIN_J15**: Connected to `KEY[0]` and `KEY[1]`.
- PIN_E1**: Connected to `KEY[1]`.
- PIN_D3**: Connected to `MIO1`.
- PIN_A15**: Connected to `LED[0]`.
- PIN_A13**: Connected to `LED[1]`.
- PIN_B13**: Connected to `LED[2]`.
- PIN_A11**: Connected to `LED[3]`.
- PIN_D1**: Connected to `LED[4]`.
- PIN_F3**: Connected to `LED[5]`.
- PIN_B1**: Connected to `LED[6]`.
- PIN_L3**: Connected to `LED[7]`.
- PIN_C3**: Connected to `MIO0`.
- PIN_A3**: Connected to `MIO3`.
- PIN_B4**: Connected to `MIO5`.
- PIN_B3**: Connected to `MIO4`.
- PIN_A2**: Connected to `MIO2`.

The circuit is powered by `VCC` and `GND`.